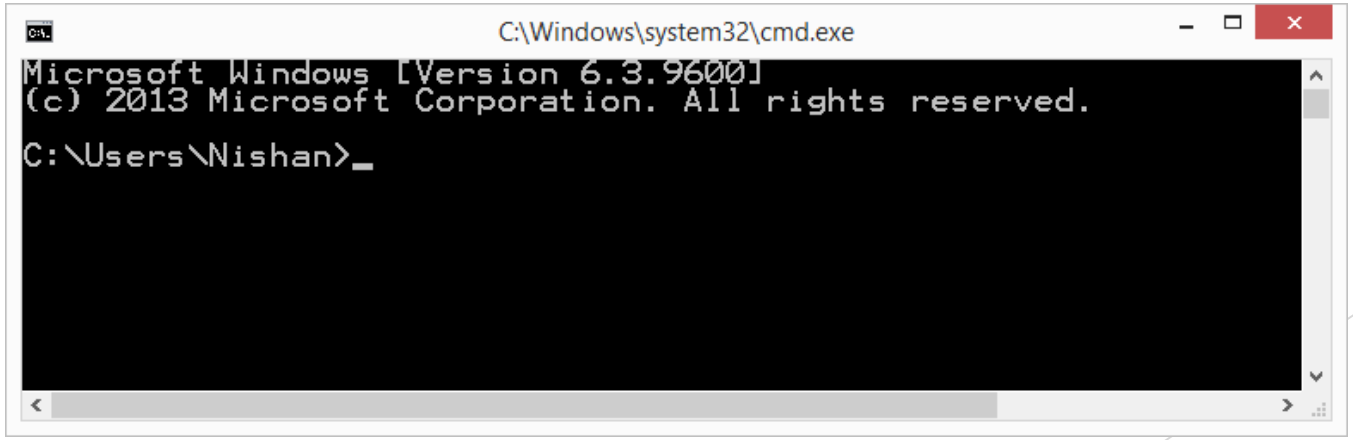


Console Arguments

Preparation Work

What Are Console Arguments?

- A file can take values from command console
- You call them as console arguments
- You can use them to do various work
  - File copying
  - Simple calculations
  - Command options, etc.



Real Life Applications 1 (Try in Console)

```
C:\Users\Nishan>vol
 Volume in drive C is Windows 8.1
 Volume Serial Number is EECA-F9CF

C:\Users\Nishan>vol d:
 Volume in drive D is Data
 Volume Serial Number is 7837-5764
```

Real Life Applications 2 (Try in Console)

```
C:\Users\Nishan>ver

Microsoft Windows [Version 6.3.9600]

C:\Users\Nishan>ver /?
Displays the Windows version.

VER
```

Real Life Applications 3 (Try in Console)

```
C:\Users\Nishan>copy Sample.txt NewFile.dat
        1 file(s) copied.
```

```
C:\Users\Nishan>copy Sample.txt NewFile.dat
        1 file(s) copied.
```

How Python Handles Console Arguments

```
import sys

print ("My File Name : ", sys.argv[0])
print ("Number of values : ", len(sys.argv))
print ("They are,\n" , str(sys.argv))
```

```
My File Name :  D:\Sample\MyPythonFile.py
Number of values :  1
They are,
 ['D:\\Sample\\MyPythonFile.py']
>>>
```

Real Console Arguments (Not IDLE)

```
import sys

print ("My File Name : ", sys.argv[0])
print ("Number of values : ", len(sys.argv))
print ("They are,\n" , str(sys.argv))
```

```
C:\>MyPythonFile.py 1 2 3 4
```

```
D:\Sample>MyPythonFile.py 1 2 3 4
My File Name :  D:\Sample\MyPythonFile.py
Number of values :  5
They are,
 ['D:\\Sample\\MyPythonFile.py', '1', '2', '3', '4']
```

Real Console Arguments (Not IDLE)

```
import sys

print ("My File Name : ", sys.argv[0])
print ("Number of values : ", len(sys.argv))
print ("They are,\n" , str(sys.argv))
```

```
C:\>MyPythonFile.py 100 500 Text.txt
```

```
D:\Sample>MyPythonFile.py 100 500 Text.txt
My File Name :  D:\Sample\MyPythonFile.py
Number of values :  4
They are,
 ['D:\\Sample\\MyPythonFile.py', '100', '500', 'Text.txt']
```

View Arguments (Another Way)

```
import sys

# Show single value
print (sys.argv[1])

# Show is a loop
for value in sys.argv:
    print (value)
```

```
C:\>ViewArgs.py 100 500 Text.txt
```

Error may come if you run this in IDLE ☹
Why ???

```
D:\Sample>ViewArgs.py hello how are you?
hello
D:\Sample\ViewArgs.py
hello
how
are
you?
```

Common Error Would Be

```
Traceback (most recent call last):
  File "D:\Sample\add.py", line 9, in <module>
    val1 = int(sys.argv[1])
IndexError: list index out of range
>>>
```

Some Useful Work

```
import sys

# Create variables
val1 = 0
val2 = 0
ans = 0

# Read values from console
val1 = int(sys.argv[1])
val2 = int(sys.argv[2])
ans = val1 + val2

# Display answer to console
print (ans)
```

```
C:\>add 5 6
```

```
D:\Sample>add 5 6
11

D:\Sample>_
```